Northeastern University

Programming Foundations for Digital Media

Final Project Report

How my university workload affects my daily habits

by

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Overview

This project began with a personal curiosity: Does the time I dedicate to my academic workload directly consume the time I have for my personal habits? Initially, my hypothesis was that as my study time increased, the time dedicated to my daily habits would decrease. I also suspected that my energy levels would be another factor that could negatively affect my routine.

As the project evolved, I realized it was important to consider my daily context, specifically, whether I had classes or not, to better understand my time management. Eventually, this led me to categorize my data into different types of days to be more precise. Additionally, I noticed that for some habits I usually spent the same amount of time on them regardless of my study load. Because of this, I slightly modified my question, instead of evaluating how much time I spent on my habits, I decided to analyze simply if my studies affected the realization of those habits.

My main goal with this project was to identify my behavioral patterns or lack of them, to create a plan to improve my time management and ensure that my academic life and personal habits can coexist in balance.

To explore this, I designed a botanical-inspired visualization that translates my daily routine into a garden, where I evaluate if study time, habit realization and type of day, have any correlation.

About the Data

I recorded my data on a daily basis from January 26th to March 10th, ensuring that each day captured my routine. My objective was to build a dataset that reflected the time spent on tasks and my perceived physical/mental state (energy levels). Then I also included the context with the type of days; this was easy to include in my dataset because I could check the calendar to confirm which days I had class, which ones were free weekdays and which ones where the weekend.

These are the types of data that I collected:

- **Date:** Recorded to maintain a chronological log of my activities. This was essential to observe if my consistency in daily habits would decrease as the quarter progressed and deadlines approached.
- **Type of Day:** This allowed me to analyze how my schedule influenced my daily routine.
    - **Class Day:** this corresponds to the days I had classes.
    - **Free Day:** weekdays that I would usually spend working on my assignments, resting or going to meetings and university events.
    - **Weekend:** days where I would usually rest or spend time with friends.

- **Studying Duration:** Measured in minutes to track the total time dedicated to academic sessions each day. I used a timer to track this information.
- **Habit Realization (Music, French, Reading, & Sleeping):** Initially, I recorded the duration of each habit in minutes using a timer; in the case of tracking the minutes I slept, I used the app BetterSleep. This was intended to help me analyze how my workload affected the specific time dedicated to each activity. Later, as the project evolved, this data served primarily to identify whether the habit was performed or not.
- **Energy Levels:** Tracked on a scale from 1 to 5, with 1 being low energy and 5 being high energy. I collected this to test the correlation between my physical/mental state and my ability to maintain my routine. This type of data was based on my judgment.
- **Time of Day:** Logged to keep a detailed track of every session, especially on days where I had multiple study or habit sessions, ensuring no data was overlooked.

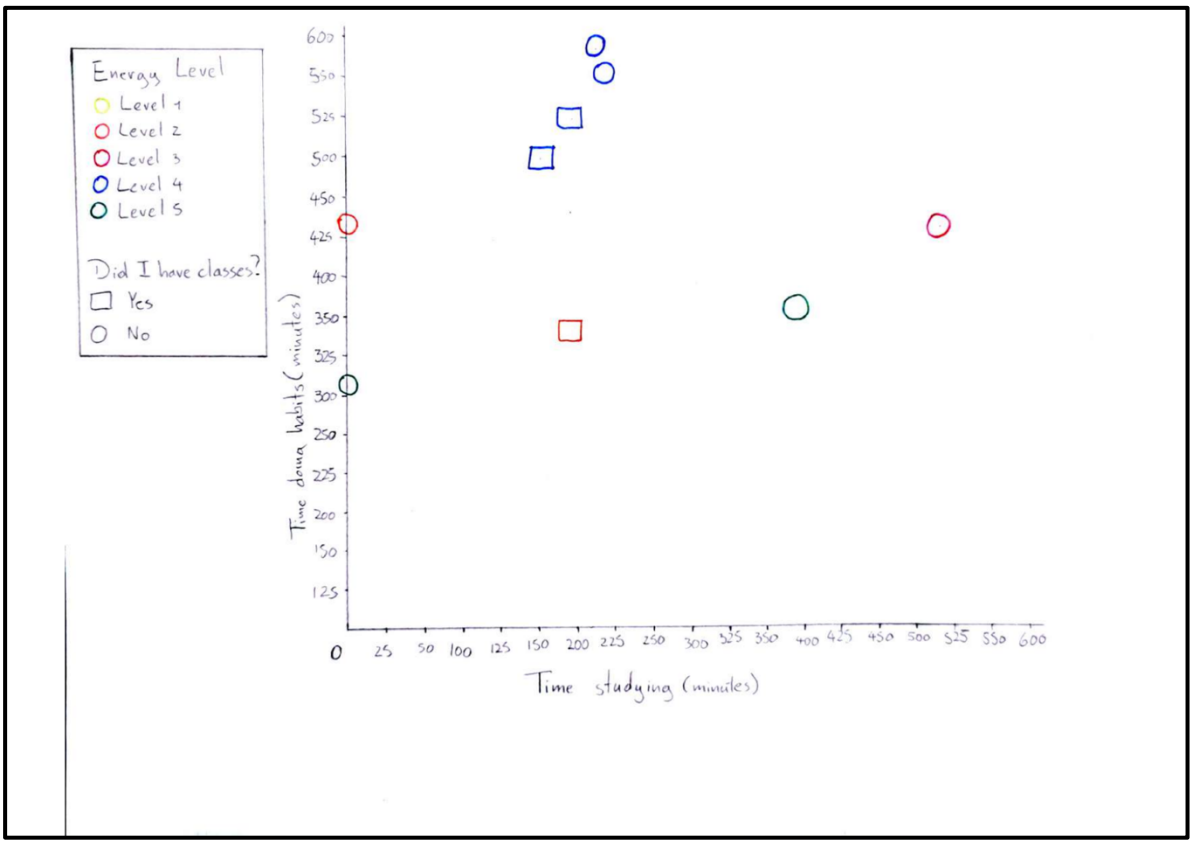
Date	Amount of time studying (min)		Amount of time Studying french (min)		Amount of time practicing music (min)		Amount of time reading (min)		Sleep Duration (min)		Level of energy (1-5)		Type of day
	Time of day	Total minutes	Time of day	Total minutes	Time of day	Total minutes	Time of day	Total minutes	Time of day	Total minutes	Time of Day	Level	
26/01/26	15:00 16:00	30 180	11:00 19:00	30 40	15:40	20	21:00	30	23:00	455	7 am	4	Free day
27/01/26	6:30 14:00	120 30	12:00:00	70	------	------	9:25	60	21:35	365	7 am	4	Class
28/01/26	8:00 15:00 22:00	300 90 120	------	------	16:30	30	------	------	23:52	405	7 am	3	Free day
29/01/26	10:00 20:00	75 120	9:08	31	------	------	------	------	12:10	310	7 am	2	Class
30/01/26	8:30 11:30 16:00 19:00	60 137 120 60	------	------	15:30	30	------	------	12:00	325	7 am	5	Free day
31/01/26	------	------	14:10	25	10:20	83	20:00	30	0:26	290	7 am	2	Weekend
01/02/26	15:00	210	11:00 12:30	32 35	14:30	30	------	------	23:00	450	7 am	4	Weekend
02/01/26	19:10	103	------	------	-------	-------	------	------	23:21	305	7 am	5	Free day
03/02/26	17:17	183	------	--------	9:40	37	------	------	22:52	488	7 am	4	Class
04/01/26	15:30 20:47	270 283	12:18	22	------	------	14:00	35	23:43	345	7 am	5	Free day
05/02/26	18:10	165	21:23	27	------	------	------	------	0:48	225	7 am	2	Class
06/01/26	9:00 14:00	120 103	19:00	38	18:08	30	16:06 17:00	21 25	23:00	290	7 am	3	Free day
07/02/26	13:50	180	10:45	20	17:00	32	21:50	26	22:32	395	7 am	5	Weekend
08/02/26	18:20	50	9:03	73	17:47	26	21:50	22	22:20	430	7 am	3	Weekend
09/02/26	17:00	220	21:00	22	15:36	15	21:57	12	22:14	440	7 am	4	Free day
10/02/26	11:25	35	22:39	37	------	------	------	------	22:10	445	7 am	5	Class
11/02/26	9:00 13:00	165 120	------	------	17:00	49	------	------	22:27	270	7 am	2	Free day
12/02/26	10:00	30	------	------	------	------	------	------	22:36	345	7 am	2	Class
13/02/26	14:30	270	10:15	36	20:10	16	------	------	23:11	380	7 am	3	Free day
14/02/26	------	------	14:19	41	------	------	------	------	3:01	315	7 am	3	Weekend
15/02/26	11:00	167	18:46	40	17:28	21	------	------	23:15	465	7 am	4	Weekend
16/02/26	16:17	67	10:39	63	17:30	15	20:04	32	21:41	455	7 am	3	Free day
17/02/26	19:30	95	9:16	32	------	------	21:22	20	20:47	515	7 am	5	Class
18/02/26	10:57 17:55	14 45	11:40	33	16:03	19	------	------	22:54	385	7 am	4	Free day
19/02/26	19:00	60	20:20	34	------	------	------	------	21:30	445	7 am	3	Class
20/02/26	10:51 15:00	49 120	12:22	80	------	------	19:20	72	23:54	395	7 am	4	Free day
21/02/26	------	------	------	------	------	------	------	------	22:27	405	7 am	4	Weekend
22/02/26	------	------	------	------	11:00	60	------	------	22:40	386	7 am	1	Weekend
23/02/26	14:00	197	17:44	30	------	------	19:20	100	23:03	445	7 am	3	Free day
24/02/26	------	------	20:10	40	------	------	21:00	30	21:30	455	7 am	4	Class
25/02/26	11:00 19:00	340 155	8:35	18	------	------	21:40	20	21:53	490	7 am	4	Free day
26/02/26	------	------	6:48	32	------	------	------	------	12:43	290	7 am	2	Class
27/02/26	------	------	18:47	62	16:15	60	20:40	46	22:47	395	7 am	4	Free day
28/02/26	------	------	11:23 18:09	66 32	11:00	47	------	------	22:00	455	7 am	4	Weekend
01/03/26	------	------	9:25	95	16:05	93	20:00	51	21:26	500	7 am	5	Weekend
02/03/26	16:00	399	9:15	91	------	------	------	------	20:51	485	7 am	4	Free day
03/03/26	9:47	124	------	------	------	------	------	------	0:07	280	7 am	2	Class
04/03/26	14:00	360	10:20	60	9:00	35	------	------	23:10	345	7 am	4	Free day
05/03/26	8:17	33	9:29	31	------	------	------	------	23:10	375	7 am	3	Class
06/03/26	------	------	10:10	108	15:00	68	6:00	43	22:17	435	7 am	4	Free day
07/03/26	10:32 13:31 17:00 21:20	120 160 180 40	20:32	35	------	------	------	------	22:35	435	7 am	4	Weekend
08/03/26	------	------	14:00	85	11:00	70	20:30	60	23:45	380	7 am	3	Weekend
09/03/26	------	------	10:00	90	15:16	44	------	-------	23:35	495	7 am	4	Free day
10/03/26	------	-------	8:00	20	10:47	19	------	-------	22:52	395	7 am	4	Class

Sketches

My final visualization was born from my “Final Project Sketches” assignment, which consisted of creating three different sketches to visualize our data in diverse ways.

First Sketch: Scatterplot

My first sketch was a scatterplot designed to analyze my initial hypothesis. This is how I initially envisioned how my final project would be. It maps out how much I study (X-Axis) versus how much time I spend on my daily habits (Y-Axis). I also wanted to evaluate if the days I had class (squares) or free days (circles), along with my energy levels (variations in color) had any influence on my routine.



Sketch 1.

I like this idea because it allows me to compare the time dedicated to my studies against the time dedicated to my habits, which is something that I can´t easily visualize in my current version, since its main focus is not time. Another aspect that I like is that I can see if my energy levels and my schedule have an influence on my daily routine. One drawback, however, is that in this sketch I am not as precise as I

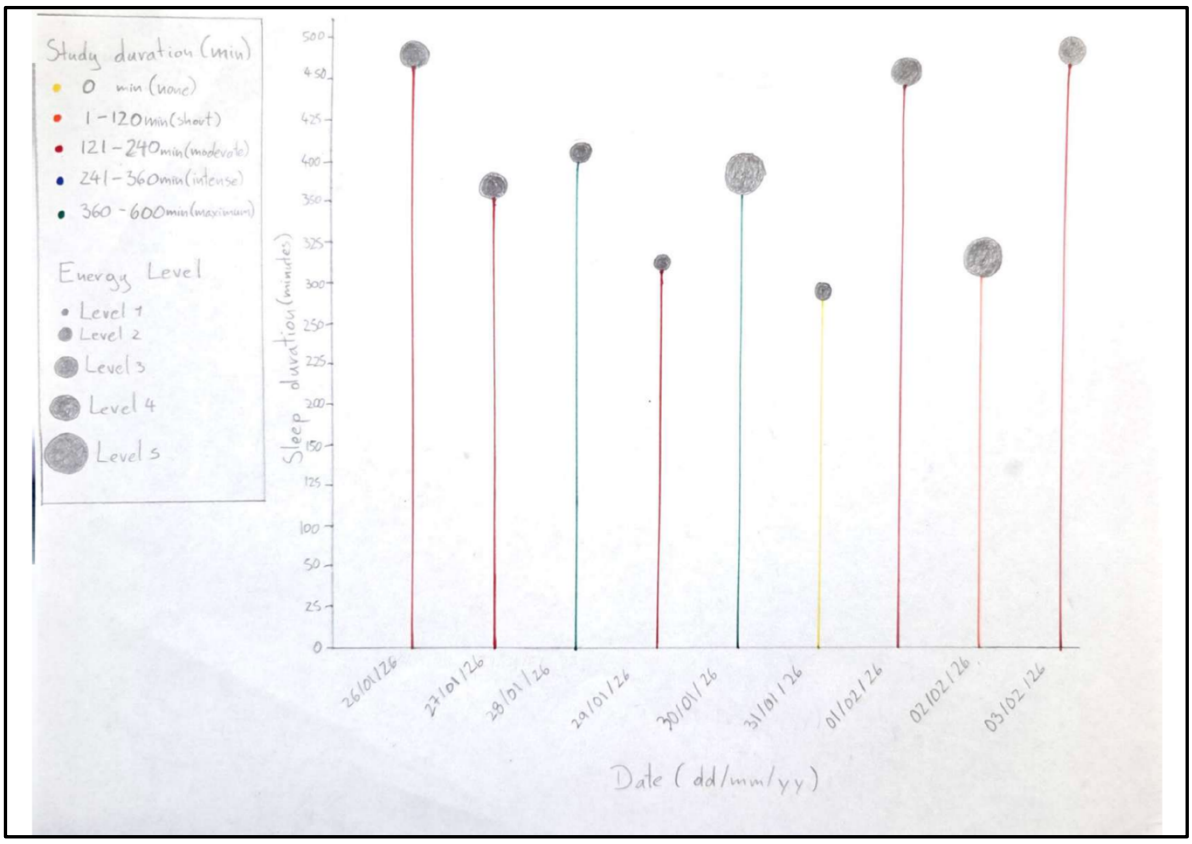
could have been regarding the types of days; at the time I created these sketches, I had not yet considered putting weekend days into their own category. Additionally, this sketch does not allow me to evaluate which specific habits I was able to perform, nor can I observe my daily routine in a clear day-to-day pattern because I can't identify the specific date corresponding to each day. This is the main reason why I decided to not use this idea.

Second Sketch: Lollipop Chart

For my second sketch I decided to change the values on my axes to create a daily report, and I also shifted my hypothesis to examine my data from another perspective and show other correlations between the data that I had been collecting.

For this second sketch, I wanted to see how my daily workload affected my sleeping habit, as that was the habit with which I was most consistent. By using a Lollipop Chart, I could track each day chronologically and determine if there was any correlation between my daily studying sessions (represented by the color of the lines, where each color represents a different time range) and my sleep duration (Y-Axis). I also included my energy levels (represented by circles attached to the colored lines; these circles have different radii depending on the energy level) to evaluate if they also affect my sleeping habit. I got the idea of using a Lollipop Chart after exploring the website *From Data to Viz*, which showcases various methods of data visualization. In my opinion the Lollipop Chart was the right choice at that stage as I wanted to include specific dates on the X-Axis to obtain a clearer idea of the daily fluctuations of my data.

The hypothesis evaluated in this chart is the following: as the time I spend studying for university increases, my sleep duration decreases. Additionally, this is negatively influenced by my energy levels.



Sketch 2.

What I highlight about this visualization is that I can see the daily routines clearly and the representation of the energy levels is improved in comparison with the previous sketch; the changing size of the circles gives me a more accurate representation of the amount of energy that I had each day. However, what is not working is the visualization of study time. Although I created different categories for various time ranges, it is not as precise as it could be.

As interesting as it was to see the relationship of just one of my habits with my study time, I decided to not pursue this idea because I still wanted to include all my habits, and I felt the color categorization of the study duration was not the best representation for that type of data.

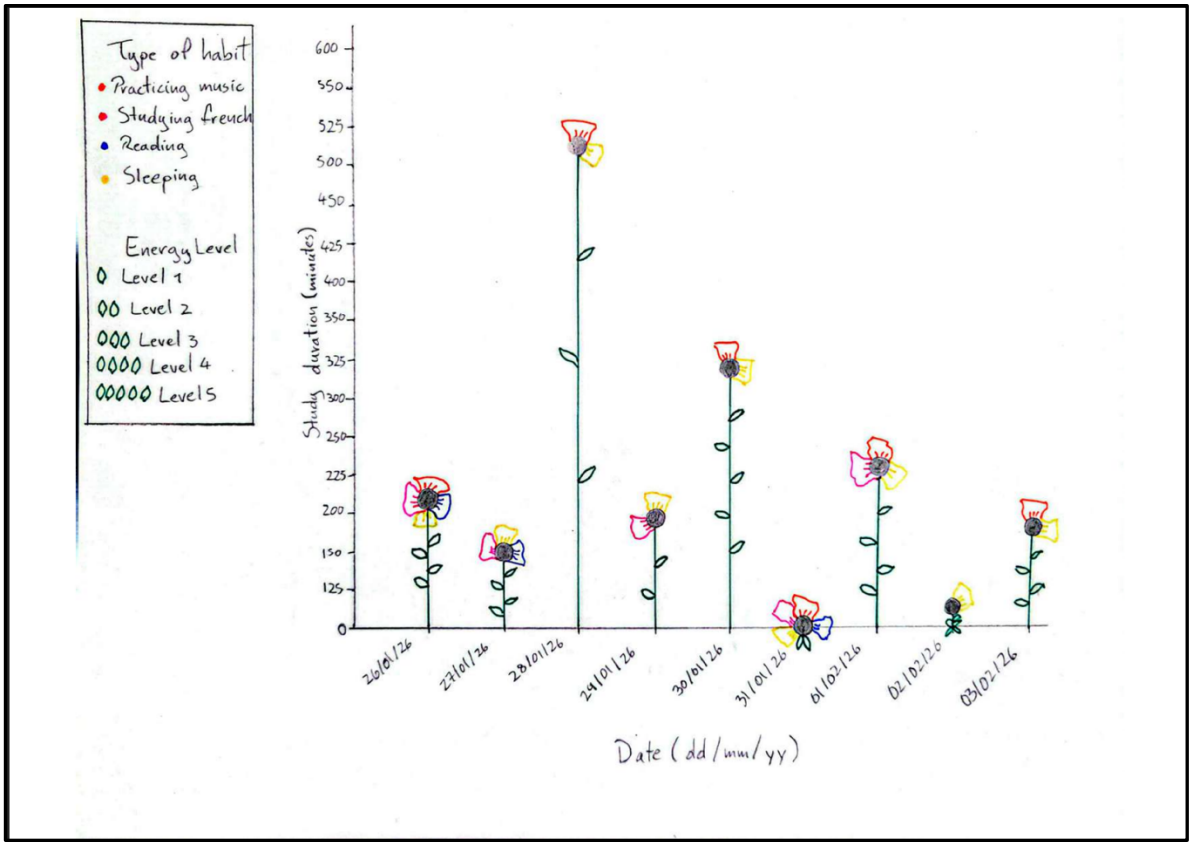
Final Visualization

For my final visualization I decided to proceed with my third sketch, which is the most creative out of the three. In this visualization, I am evaluating how my workload and energy levels affect the consistency of my daily habits. I developed this idea as an evolution of my previous two sketches; from the scatterplot I decided which types of

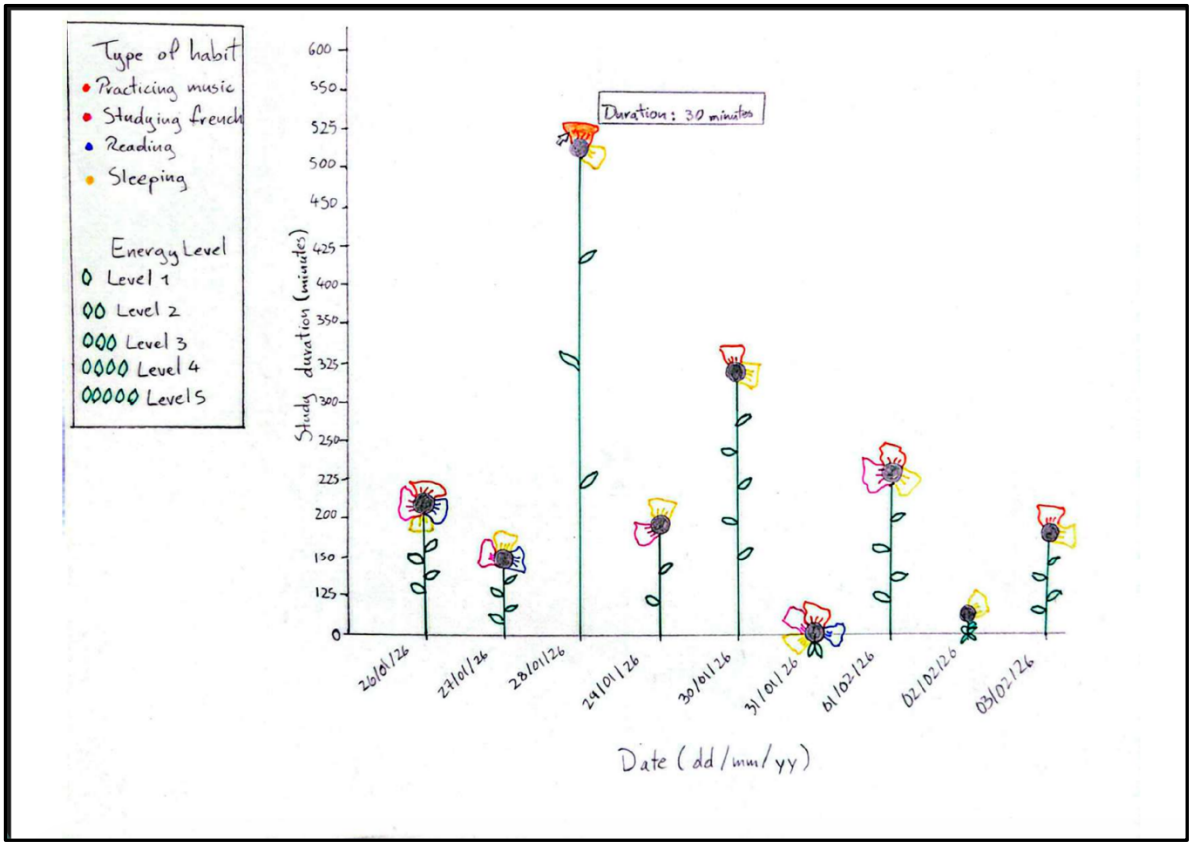
data to visualize, and the idea for the flowers came from the Lollipop Chart. I noticed that a line with a circle at the top resembled a flower without petals, and then I remembered some examples from the *Dear Data* website where nature was used as a form of visualization.

In this sketch, each flower represents a day: the length of the stem represents the study duration, the number of leaves indicates my energy level, and each petal represents a specific habit, with its color indicating the category. This serves as a visual representation of how well I managed my routine even when the workload was significant or my energy was low; it also shows which habits I performed more frequently. Additionally, I also wanted to include an interaction to display the exact duration of a habit in minutes when the mouse hovered over a petal.

With these changes my new hypothesis became the following: as the time I spend studying for university increases, the number of my habits performed daily decreases. In addition, this is negatively influenced by my energy levels.



Sketch 3 without interaction.



Sketch 3 with interaction.

I chose this visualization for multiple reasons. The first was that I wanted to do something different from a scatterplot and challenge myself creatively and technically. The second reason is that, as I mentioned before, as I was collecting my data, I noticed that for certain habits I usually spent a similar amount of time doing them, regardless of whether I had classes or not. Therefore, I considered it better to change my perspective and evaluate if I was performing my habits instead of just comparing durations. The third reason is related to my main goal for this project, identifying my behavioral patterns or lack of them, to create a plan to improve my time management and ensure that my academic life and personal habits can coexist in balance. With my initial representation I could see how much time I was dedicating to my habits, but I couldn't truly identify how each one was affected individually. I also couldn't properly identify how my schedule affected my habits because I did not have a clear chronological visualization. This final visualization is the one that helps me the most to evaluate my daily routine, create a plan for improvement and achieve a better time management.

The only two differences between my original sketch and the final result are that I included the data of the types of days by using a specific shape for each one, and

that I decided not to visualize the energy levels as leaves. The latter was a design decision because as you can see in the sketch, for the days in which I did not study, there is not stem for me to position them. By putting them alongside the petals, the visualization turns messy, so I decided to include my energy levels in the interaction's pop-up box with the duration of the habits

One downside with my visualization is that it requires horizontal scrolling to see all the data inputs and, by doing this you lose sight of the Y-Axis. I tried to fix this by scaling the visualization, but it did not work because it compromised the readability of the data, the shapes, numbering and labeling ended up being too small. My solution was to create another Y-Axis on the right so that the viewer could still see its values and make a proper reading of the visualization. I also made sure to leave the key in an adequate position so the viewer could always see it.

Self-Analysis

From the beginning of the project, I knew that I wanted to explore the relationship of my study duration and my habits. I always thought that my workload was the primary reason I often couldn't maintain the routines I wanted; I was aware that I tended to prioritize my studies over my personal interests. This project offered the perfect opportunity to prove whether this hypothesis was correct, especially since I had been consistent with my habits for two weeks before starting the quarter when I had more free time. It was interesting to see how the return to classes would affect that consistency.

After analyzing the final visualization, I have disproven my initial hypothesis. The data shows no direct correlation between the time I spend studying and the realization of my habits. To my surprise, some days I was able to complete all my habits despite having long study sessions. Also, on other days when I had no classes or study sessions, I still barely performed any of my habits.

Including the "type of day" in the visualization was useful because it showed that while class days were naturally more difficult for maintaining a routine, the results on free days were what truly disproved my theory. Some of my most productive study days still featured three petals (habits), while weekends showed high variability, with habits often disappearing regardless of the workload.

Regarding the influence of my energy levels, while most days maintain a high energy level of 4 or 5, they don't show a proportional reflection on my habits' productivity. Some days with high energy levels and low study time, have few petals on the

flowers. On the other hand, the data also shows that whenever my energy drops to 2, the flower remains incomplete regardless of whether the study load was heavy or light. This suggests that a level 2 energy marks a "threshold" where my personal habits routine starts to break down but for the most part, I'm still consistent with my study. There is only one exception for this with January 31st, because I did manage to perform all my habits.

Seeing my process in retrospective, I realize that my habits could be driven by variables that I did not consider for my data collection such as motivation levels, the presence of important academic deadlines, or the time I spend on my smartphone. For example, I recall that even on days when my energy level was at a 5, I sometimes failed to perform my habits due to a lack of motivation and instead I decided it was best to watch a series. To truly understand what affects my routine, a future iteration of this project would need to include those other factors.

Ultimately, this project revealed that my workload is not the main obstacle, but rather a complex combination of different factors. In addition, the data suggests there could be an issue with time management and/or a lack of a defined daily structure.

Before being assigned this project, I was already collecting data on my daily habits and the most relevant event of each day. I started doing this on January 1st, 2026, without knowing that this project would eventually exist. My method of data collection was less detailed, I simply indicated whether or not I did my daily habits and wrote down the most significant thing that happened each day. I began this practice to work on my discipline and to have a tangible representation of my progress.

Until the start of this project, I had not considered on collecting other types of data to see what factors might influence my habits. I found the process incredibly useful, and I plan to apply these new data collection methods to my existing routine to better understand how it works. Additionally, this experience has made me realize the importance of tracking my sleeping habits. One of the key takeaways from my project, besides improving my time management and working on a daily structure, is the clear need to improve my sleep schedule in order to enhance my overall sleep quality.

Bibliography

Holtz, Yan, and Connor Healy. "Find the Graphic You Need". *From Data to Viz,* 2018, www.data-to-viz.com/.

Lupi, Giorgia, and Stefanie Posavec. “The Project”. *Dear Data*, 2016, www.dear-data.com/theproject.